

Class 4:

Evaluate the following integrals by changing to polar coordinates:

1. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{(1+x^2+y^2)^{3/2}} dx dy$

ans: 2π

2. $\iint y^2 dx dy$ over the area outside $x^2 + y^2 - ax = 0$ and inside $x^2 + y^2 - 2ax = 0$ ans: $\frac{15\pi a^4}{64}$

3. $\iint \frac{\sqrt{1-x^2-y^2}}{1+x^2+y^2} dx dy$ the integral being extended over all positive values of x and y subject to $x^2 + y^2 \leq 1$.

ans: $\frac{\pi^2}{8} - \frac{\pi}{4}$
